9012 - The missing link for understanding multiple populations in star clusters

Cycle: 4, Proposal Category: GO

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1		NIRCam Imaging	(1) NGC-1846
	2	Residual Persistence Check	NIRCam Dark	NONE

ABSTRACT

All massive Globular Clusters (GCs) older than ~2 Gyr host multiple populations (MPs) of stars with different chemical compositions. Understanding the origin of this phenomenon is a challenge in modern astrophysics.

One scenario is that GCs were substantially more massive at formation, had several episodes of star formation, and the subsequent mass loss provided a significant contribution to the early assembly of the Galaxy. Alternatively, GCs consist of only one stellar generation, and the MPs are the result of mass accretion onto pre-existing stars.

MPs have never been observed in LMC clusters younger than 2 Gyr, where stars heavier than 1.5 solar masses (the only ones that can be studied without JWST) are chemically homogeneous. Nevertheless, these clusters host groups of stars with different rotation rates and masses larger than ~1.5 solar masses.

The fact that the age boundary at which MPs are present is also the boundary at which the effects of rotation disappear (i.e. where stars are magnetically braked) suggests that MPs do not correspond to multiple generations but are instead caused by a non-standard stellar evolutionary effect linked to rotation.

We propose deep NIRCam imaging of the massive ~1.6 Gyr-old GC NGC1846, which does not host MPs among stars havier than ~1.5 solar masses, and explore for the first time its lowest-mass stars (M-dwarfs). Will they show the MPs expected if the low-mass stars have different chemical compositions as in the old GCs?

Or will their low MS remain narrow as expected for a simple generation?

The outcome will provide us with the link for understanding the formation of massive clusters in the present day and in the distant past.

OBSERVING DESCRIPTION

We propose to collect images through the F115W and F322W2 filters of the NIRCam detector of low-mass stars in a young (1.6 Gyr old) and massive globular cluster in the Large Magellanic Cloud.

The main goals of our project require high-precision photometry and astrometry of M-dwarfs in a relatively large field of view. This dataset will be used to identify and analyze multiple stellar populations among the low-mass stars of our target if present. It will allow us to shed light on the origin of globular clusters and their stellar populations and on the role of globular clusters in the host-galaxy assembly.

To do this, we ask for a large number of deep and well-dithered images that will allow us to measure thousands of bright stars in different regions of the detector. Such a dither strategy will allow us to derive high-precision maps of the geometric distortion of the detectors and accurate models of the point-spread function and of its spatial variation. The NIRCam is the optimal camera for our purposes because

i) it provides the widest field of view and

ii)allows simultaneous observations in the F115W and F322W2 bands. As demonstrated in the proposal, these are the most-efficient filters to identify and characterize multiple populations of low-mass stars.

iii) Thanks to its high spatial resolution, NIRCam is the optimal camera to get high-precision photometry in the crowded environment of an LMC star cluster.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	NGC-1846	RA: 05 07 42.9000 (76.9287500d) Dec: -67 28 28.45 (-67.47457d) Equinox: J2000	Epoch of Position: 2015.5	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Stellar Cluster
Description=[Globular star clusters]

Observation

Proposal 9012, Observation 1

Thu Sep 18 22:00:18 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Imaging

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:4) Warning (Form): Overheads are provisional until the Visit Planner has been run.
(Visit 1:5) Warning (Form): Overheads are provisional until the Visit Planner has been run.

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Template

Module	Subarray	Target Placement
ALL	FULL	Module Gap

Dithers

#	Primary Dither Type	Primary Dithers	Subpixel Dither Type	Dither Size	Subpixel Positions
1	FULL	36	STANDARD		1

Spectral Elements

#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID
1	F115W	F322W2	DEEP8	7	1	36	36	49475.036	

Special Requirements

Sequence Visits , Non-interruptible
Aperture PA Range 305 to 325 Degrees (V3 305.07457694 to 325.07457694)
Visits Same PA
Persistence Risk

2 After 1 by 26.6 Hours to 2 Days, Exclusive Use Of Instrument

Observation

Proposal 9012, Observation 2: Residual Persistence Check

Thu Sep 18 22:00:18 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRCam Dark

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Template

Module	Subarray	Science Template	Occulting Mask	No. of Output Channels
ALL	FULL	Imaging		4

Spectral Elements

#	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Total Integrations	Total Exposure Time	Optional ETC ID
1	MEDIUM8	10	1	1	1	1052.203	

Special Requirements

2 After 1 by 26.6 Hours to 2 Days, Exclusive Use Of Instrument